

Brazilian Review of Finance: A Template

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Abstract Your abstract goes here! Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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1. Introduction

In the era of digital technology, where information is seamlessly transmitted worldwide, the foundation of the interconnected Internet of Things (IoT) is formed by complex network protocols ensuring the reliability and interoperability of communication networks. To safeguard these pervasive ecosystems, security protocols rely fundamentally on cryptographic algorithms to ensure data integrity, confidentiality, mutual authentication, and non-repudiation.

At the core of these cryptographic guarantees are Authenticated Key Exchange (AKE) protocols, which serve as the fundamental mechanisms for establishing secure channels over untrusted networks. The primary objective of an AKE protocol is to enable two communicating entities to dynamically

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negotiate a shared cryptographic session key while mutually verifying their identities. This dual functionality is critical for thwarting active adversaries, particularly those attempting Man-in-the-Middle (MitM) attacks. Over the past decades, the cryptographic community has developed and standardized several classic AKE protocols. Prominent examples include the foundational Station-to-Station (STS) protocol[5], the SIGMA (SIGn-and-MAC) family[7], and the robust key exchange frameworks underpinning ubiquitous industry standards such as Transport Layer Security (TLS)[4] and the Internet Key Exchange (IKE)[6].

Beyond theoretical cryptographic guarantees, the practical deployment of AKE protocols in the highly fragmented IoT landscape has increasingly highlighted the critical need for cross-device and cross-vendor interoperability. Much like how TLS evolved to become the ubiquitous, unifying standard for securing web traffic, there is a strong industry-wide drive to consolidate disparate IoT communication frameworks into universal standards. Spearheading this unification effort is the Connectivity Standards Alliance (CSA), which successfully introduced the Matter protocol[2]—a landmark standard that establishes a seamless, interoperable security and application layer for smart home ecosystems. Building upon the robust AKE frameworks and interoperability principles proven by Matter, the industry is now extending these unification efforts into more security-critical physical domains, particularly digital access control and contactless authentication across heterogeneous mobile platforms.

To address this critical gap in physical security, the CSA has spearheaded the development of Aliro[1], a dedicated standard designed to revolutionize digital access experiences. Operating alongside Matter as a complementary framework within the CSA ecosystem, Aliro aims to establish a universally interoperable architecture for smart locks and digital credentials across diverse mobile devices and wearables. At the core of Aliro's security model is a stringent reliance on AKE mechanisms tailored for proximity and short-range wireless environments, predominantly Near-Field Communication (NFC) and Bluetooth Low Energy (BLE). By leveraging classic cryptographic primitives—specifically Elliptic Curve Cryptography (ECC) and Elliptic Curve Diffie-Hellman (ECDH)—the Aliro protocol facilitates secure, offline credential verification and mutual authentication between a user's terminal and an access reader across these constrained wireless channels.

However, the long-term viability of ECC-based access control frameworks like Aliro is facing an existential threat from the rapid advancement of quantum computing. Recent studies indicate that the imminent advent of

Cryptographically Relevant Quantum Computers (CRQCs) will render Shor's algorithm practically executable, enabling the resolution of the Elliptic Curve Discrete Logarithm Problem (ECDLP) in polynomial time[3][8][9]. Once realized, CRQCs will completely compromise the ECDH key exchanges and ECDSA signatures that underpin Aliro's security model. This mathematical vulnerability would theoretically allow adversaries to forge offline access credentials in real-time or execute 'harvest now, decrypt later' attacks, effectively dismantling the security of critical physical infrastructures. In response to this impending cryptographic crisis, there is an urgent mandate to transition the IoT and access control industries toward Post-Quantum Cryptography (PQC), leveraging NIST-standardized, quantum-resistant algorithms—such as ML-KEM and ML-DSA—to future-proof these ubiquitous digital systems.

1.1 Related work

In the realm of IoT, particularly within physical access control systems, protocol design is fundamentally constrained by the severe hardware limitations of edge devices. Smart locks and contactless digital credentials typically operate under strict power budgets, limited memory footprints, and restricted computational capacities. Furthermore, these systems predominantly rely on short-range wireless interfaces, primarily NFC and BLE, which are characterized by constrained data transmission rates (e.g., NFC typically operating at 106 to 424 kbps, and BLE facing similar practical application-layer throughput limits) and stringent real-time requirements, demanding mutual authentication to be completed within a user-perceptible threshold of a few hundred milliseconds. To navigate the delicate trade-off between robust security and operational efficiency under these constraints, modern AKE protocols have overwhelmingly adopted ECC. A prime example of this paradigm is the classic Aliro protocol, which leverages the compact key sizes and high computational efficiency of primitives like ECDH and ECDSA to facilitate secure, offline authentication. In this practical framework, users employ mobile devices—such as smartphones and smartwatches equipped with NFC and BLE capabilities—to interact directly with smart locks for contactless access control. By relying on these lightweight cryptographic mechanisms, Aliro achieves seamless, millisecond-level mutual authentication that remains entirely imperceptible to the user, effectively satisfying the extreme physical limitations of the underlying hardware.

The imperative migration toward Post-Quantum Cryptography (PQC) fundamentally disrupts the established operational equilibrium of ECC-based access systems. Integrating quantum-resistant primitives into IoT protocols

introduces profound engineering bottlenecks, primarily driven by the drastic inflation of communication overhead. Unlike the compact nature of ECC, PQC algorithms inherently exhibit substantially larger public key, ciphertext, and signature sizes, which directly conflict with the narrow bandwidth and strict latency thresholds of both NFC and BLE channels. To navigate this intrinsic 'efficiency versus space' dilemma, the cryptographic community and standardization bodies, most notably NIST, have converged on lattice-based cryptography as the optimal architectural foundation. Specifically, the recently standardized ML-KEM (derived from CRYSTALS-Kyber) and ML-DSA (derived from CRYSTALS-Dilithium) have been widely adopted for key encapsulation and digital signatures, respectively. The cryptographic hardness of ML-KEM is rigorously reduced to the Module Learning with Errors (MLWE) problem, whereas ML-DSA derives its security from both the MLWE and the Module Short Integer Solution (MSIS) problems. By offering a highly optimized balance between cryptographic payload dimensions and algorithmic execution speed, these lattice-based schemes represent the most viable pathway for future-proofing resource-constrained ecosystems, albeit necessitating meticulous protocol-level redesigns to accommodate their expanded transmission footprints.

1.2 Our contribution

2. Preliminaries

2.1 Authenticated key exchange

2.2 Mathematical background

2.3 Android security architecture and hardware roots of trust

3. The PQ-Aliro protocol

4. Security analysis

4.1 Security modle

4.2 Security properties

5. Implementation and optimizing design

5.1 Secury implementation on Andriod platform

5.2 Faster Noen on cortex

6. Conclusion

7. Acknowledgement

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